Education and Information Technologies (2025) 30:4607–4638
https://doi.org/10.1007/s10639-024-12963-x

Development and validation of the perceived interactivity of learner-AI interaction scale

Feifei Wang[1] · **Alan C. K. Cheung**[1] · **Ching Sing Chai**[2] · **Jin Liu**[3]

Received: 28 March 2024 / Accepted: 7 August 2024 / Published online: 31 August 2024
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Abstract

As learners are able to perceive interactivity when interacting with instructors or peer learners in traditional learning environments, learners are similarly able to perceive interactivity when interacting with artificial intelligence (AI) in AI-supported learning environments. Advancements in AI, such as generative AI including ChatGPT and Midjourney, enhance learners' perceived interactivity, thereby facilitating learning through AI-enabled interaction. However, there is no scale in education for measuring perceived interactivity of learner-AI interaction. This study develops a 17-item scale to assess the extent to which a learner perceives interactivity with AI from four dimensions: responsiveness, personalization, learner control, and learning engagement. The sample group included 422 Chinese university students for the first application and 306 university students for the second application. Both the exploratory factor analysis and the confirmatory factor analysis verified the factor structure of the scale. The Cronbach's alpha value for the whole scale was 0.948, whereas the Cronbach's alpha values for the four dimensions ranged between 0.820 and 0.915. Results suggested that this scale was a reliable and valid instrument. This study also found that perceived interactivity of learner-AI interaction was significantly associated with AI tools, learners' behavioral intentions to use AI in learning, months of using AI in learning, and average duration of using AI in learning each time, and not associated with ages, genders, education levels, and fields of education. Finally, theoretical and practical implications are discussed.

Keywords Perceived interactivity · Artificial intelligence · Validity · Reliability · Scale development

Interactivity has long been considered crucial for knowledge acquisition, learning engagement, and cognitive skill development (Blasco-Arcas et al., 2013; Kim &

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Kwon, 2024; Wagner, 1997; Wang et al., 2023b). However, this interdisciplinary concept has mostly been investigated in fields other than education, such as business (Gao et al., 2010; Liu, 2003), computer science (Borsci et al., 2021), and communication and media (Sohn & Lee, 2005). After Sims (1997) proposed the question, "Is interactivity a forgotten art?" in education, an increasing number of educational researchers turned their attention to this art (Kettanurak et al., 2001; Kwastek, 2008). The research in education mainly focused on interactivity in learner-instructor interaction, learner-learner interaction, learner-content interaction, and learner-self interaction (Sun & Hsu, 2013). Meanwhile, the advancement of artificial intelligence (AI) has seen more and more technological developments, such as generative AI. Generative AI relies on deep learning models to produce diverse simulated interactions with learners, as exemplified by ChatGPT for generating human-like conversations and Midjourney for creating text-based images (Chien et al., 2024). With the popularity of AI tools and advancements in generative AI, a recent surge of research on learner-AI interaction has highlighted opportunities for traditional learning environments, showcasing AI's potential to provide unlimited flexibility in space and time, personalized learning experiences, and increased interactivity (Chiu, 2023; Zhai et al., 2021). Thus, interactivity in learner-AI interaction has become a noteworthy topic in AI-supported learning environments (Chiu, 2023; Memarian & Doleck, 2024; Petersen et al., 2022).

Research indicates that increased interactivity in learner-AI interaction means offering learners more choices, control, and feedback, thus producing better learning outcomes (Patwardhan & Murthy, 2015; Wang et al., 2023c). However, "perceived interactivity", an under-researched concept, is far more important than "interactivity" in learner-AI interaction (Sun & Hsu, 2013). Less is known about perceived interactivity of learner-AI interaction. Current research on perceived interactivity of learner-AI interaction has suggested two further research avenues. First, given that no scale in education has been designed to measure perceived interactivity of learner-AI interaction, a scale on this noteworthy topic is needed to help researchers, educators, AI developers, and other relevant stakeholders to identify the relationship between perceived interactivity and other variables that influence learners' perceptions, behaviors, and achievement when learners use AI in learning (Shin et al., 2016). Perceived interactivity is likely to be associated with variables such as behavioral intention to use, perceived ease of use, perceived usefulness, attitude, actual use, self-efficacy, trust, collaborative learning, and learning performance (Liu & Ma, 2023; Patwardhan & Murthy, 2015). Second, since there remains a lack of consensus on the definitions of interactivity and perceived interactivity, the inconsistent usage of the definitions in empirical studies complicates the comparability of relevant studies (Petersen et al., 2022). Therefore, there is a need for such a scale to measure perceived interactivity of learner-AI interaction, based on clear definitions of interactivity and perceived interactivity that are appropriate for AI-supported learning environments.

To address these needs for future research, this study aims to develop and validate the Perceived Interactivity of Learner-AI Interaction Scale to measure learners' perceived interactivity when interacting with AI in learning. To this end, this study takes the following steps of scale development and validation: (1) identifying clear definitions of interactivity and perceived interactivity in AI-supported learning

environments, (2) constructing a theoretical framework of perceived interactivity of learner-AI interaction, (3) developing and validating the scale, and (4) testing the relationship between perceived interactivity of learner-AI interaction and other relevant personal and AI usage factors (e.g., behavioral intentions to use AI in learning and demographic characteristics).

1 Literature review

AI expands the learning style applied in traditional education by offering advantages such as personalized learning, immediate feedback, and unlimited flexibility in space and time within interactive learning environments (Zhai et al., 2021). With the increasing accessibility and popularity of AI tools, especially advancements in generative AI, learners are experiencing enhanced interactivity with AI in learning (Chiu, 2023). A recent surge of research on the under-researched topic of perceived interactivity of learner-AI interaction has provided new opportunities and challenges; thus, perceived interactivity of learner-AI interaction has become a crucial issue in AI-supported learning environments (Memarian & Doleck, 2024).

Current research indicates that perceived interactivity of learner-AI interaction demonstrates considerable potential compared to traditional forms of learning. Some studies highlight that perceived interactivity is a key feature of interactive educational activities that enables learners to acquire knowledge while interacting with AI (Chien et al., 2024; Koć-Januchta et al., 2020). Perceived interactivity of learner-AI interaction acts as a scaffolding mechanism that not only enhances learners' engagement and reduces anxiety during activities, but also facilitates the flow status of the learning process (Wang et al., 2023b). Other studies indicate that high levels of perceived interactivity of learner-AI interaction are associated with positive learning effects for offering learners more choices, control, and feedback (Chiu, 2023; Wang et al., 2023c).

Despite the considerable potential of perceived interactivity of learner-AI interaction, there has been a challenge in research on this topic. The challenge is that the two concepts, interactivity and perceived interactivity, have been used interchangeably due to a lack of consensus on their definitions (Girish et al., 2022; Petersen et al., 2022; Wang et al., 2023a). To address the challenge, the following sections of the literature review will present three parts: an overview of research related to the definitions, a theoretical framework for perceived interactivity of learner-AI interaction conceptualized from the literature, and an exploration of the literature regarding personal and AI usage factors potentially associated with perceived interactivity of learner-AI interaction.

1.1 Definitions of interactivity and perceived interactivity of learner-AI interaction

In AI-supported learning environments, interaction, interactivity, and perceived interactivity are similar concepts that are often used interchangeably, but there are distinctions worth noting among them (Petersen et al., 2022; Wagner, 1997). Interaction

refers to the process in which two objects or events reciprocally influence each other through synchronous and/or asynchronous communication carried out by means of responses, feedback, or technology (Koć-Januchta et al., 2020; Wagner, 1997). Interactivity has been defined as the connection between subsequent messages and the extent to which earlier messages refer to even preceding messages in a series of communication exchanges (Petersen et al., 2022; Rafaeli, 1988). Perceived interactivity is the perception that learners have about their control over the interaction process or about responses from their communicative partners regarding their communicative behaviors in meaningful and engaging interaction (Shao & Chen, 2021; Sims, 1997). In this sense, interactivity emphasizes real and observable communication exchanges in interaction and thus implies real human behavior (Petersen et al., 2022), or systems' functional features and actions (Plass & Pawar, 2020). Perceived interactivity is subjectively experienced by learners and can be measured by asking learners their perceptions during the use of technology (Shao & Chen, 2021). Detailed distinctions of interaction, interactivity, and perceived interactivity are shown in Table 1, including their definitions, key attributes, measurements, and examples.

The relationship between interactivity and perceived interactivity, as actual system function/real learner behavior and perception, is complex. Perceived interactivity can be high even when only a single valued feature of technology is available, and can be low even when many interactivity features of technology are present if participants do not use them or do not find them helpful (Rodríguez-Ardura & Meseguer-Artola, 2016). While interactivity is necessary in interaction, perceived interactivity is more

Table 1 Distinctions of interactivity, perceived interactivity, and interaction in AI-supported learning environments

Category	Interactivity	Perceived interactivity	Interaction
Definition	The connection between subsequent messages and the extent to which earlier messages refer to even preceding messages in a series of communication exchanges (Petersen et al., 2022; Rafaeli, 1988)	The perception that learners have about their control over the interaction process or about responses from their communicative partners regarding their communicative behaviors in meaningful and engaging interaction (Shao & Chen, 2021; Sims, 1997)	The process in which two objects or events reciprocally influence each other through synchronous and/or asynchronous communication carried out by means of responses, feedback, or technology (Koć-Januchta et al., 2020; Wagner, 1997)
Key attribute	Objective, actual system function, real and observable learner behavior	Subjective, personal thoughts or beliefs	Objective and/or subjective, process
Measurement	Log data, number of points and clicks, learning record, etc.	Questionnaires, interviews, etc.	Usage statistics, questionnaires, interviews, etc.
Example	The inherent function of platforms that encourage dialogue and collective exploration (Alam & Mohanty, 2023)	E-learners' perception of their communication in the virtual learning environment as being bi-directional, controllable, and responsive to their actions (Rodríguez-Ardura & Meseguer-Artola, 2016)	The learning process that students communicate with AI to learn language with AI's guidance and feedback (Wang et al., 2023c)

important than actuality of interactivity when learners interact with technology (Shin et al., 2016; Sun & Hsu, 2013).

Perceived interactivity has attracted attention in research on AI-supported learning environments. With advancements in AI's interactive capabilities and proliferation of AI tools in education, learner-AI interaction has been generated to improve such learners' abilities as creativity, critical thinking, and self-regulation that can affect their performances and perceptions (Zhai et al., 2021). Perceived interactivity of learner-AI interaction has shown considerable potential compared to traditional ways of learning. Higher levels of perceived interactivity are associated with better instructional effects and enhanced learning achievement (Wang et al., 2023a; Zhang et al., 2021). Moreover, perceived interactivity helps to improve learners' perceived enjoyment, participation intention, satisfaction, and perceived learning (Benvenuti et al., 2023; Wang & Cheung, 2024).

1.2 A theoretical framework of perceived interactivity of learner-AI interaction

After clarifying and establishing the definition of the concept, the next important step in developing a reliable measure is to identify its theoretical dimensions (Churchill, 1979; Yim et al., 2023). According to previous studies that have analyzed perceived interactivity and characteristics of learner-AI interaction, perceived interactivity of learner-AI interaction is grounded in constructivist principles, where learners assimilate new knowledge by connecting it with their existing understanding and actively engaging in the co-construction of knowledge with AI (Robertson et al., 2024; Wang et al., 2023c). The interaction process consists of AI-initiated and learner-initiated interaction, with learner engagement as the main component to reflect the active involvement of learners (Alanne, 2021). Therefore, the Engagement Control Model (Sims, 1997) is deemed appropriate to be the conceptual basis of the theoretical framework of perceived interactivity of learner-AI interaction, for this model is grounded in constructivism with emphasis on learners' engagement, involvement, and motivation in interaction with technology. This model proposes two key dimensions of the concept regarding learner-initiated interaction: learner control and learning engagement. Two other dimensions that are frequently highlighted in the literature include responsiveness and personalization (Alimamy & Gnoth, 2022; Gao et al., 2010; Liu, 2003; Xu et al., 2011), both of which fall under AI-initiated interaction. Taken together, the current body of research indicates that the theoretical framework of perceived interactivity of learner-AI interaction consists of the four dimensions, as shown in Fig. 1.

Specifically, in the framework, learning engagement and learner control are typically associated with learner-initiated interaction (Sims, 1997), and responsiveness and personalization are generally associated with AI-initiated interaction (Chandra & Rahman, 2024). In situations where restrictions of interaction (e.g., time, place, and skill levels in using AI) regarding learner control and responsiveness are minimized, higher levels of learning engagement and personalization may be enhanced (Lee, 2005; Xue et al., 2023).

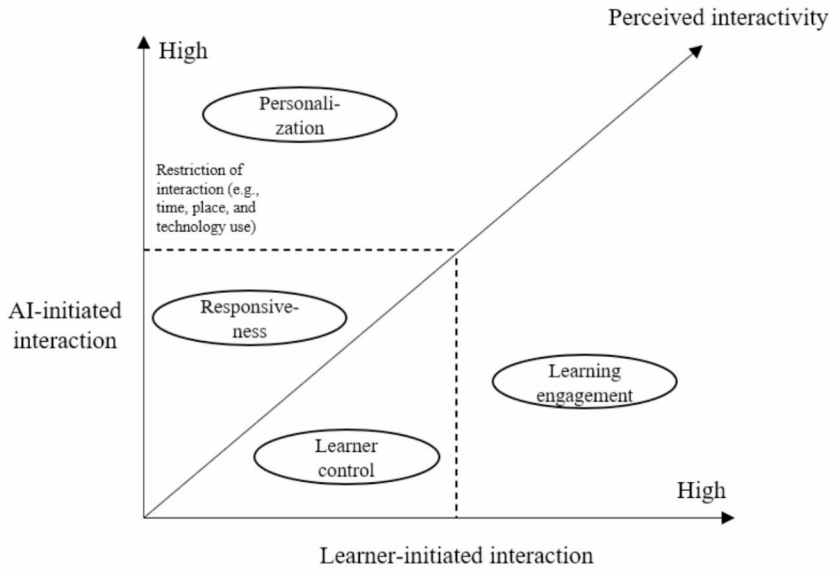


Fig. 1 The Framework of Perceived Interactivity of Learner-AI Interaction

1.2.1 Responsiveness

Many researchers emphasized the importance of responsiveness in the concept of perceived interactivity (e.g., Shao & Chen, 2021; Voorveld et al., 2011). Responsiveness refers to interactive communication with fast, appropriate, and relevant messages to sustain a thread of continuity (Johnson et al., 2006). When a learner perceives that the temporal delay between messages decreases, and that the appropriateness and relevance between messages increase, the coupling of learner-AI interaction becomes closer and more intense, so interactivity grows (Walther, 2020). In learner-AI interaction, a message is perceived as highly responsive if it is fast, appropriate, and relevant, and maintains the continuity of interaction.

1.2.2 Personalization

Personalization was found to be a saliant factor of perceived interactivity (Plass & Pawar, 2020; Song & Zinkhan, 2008). Personalization has been defined as the ability to provide learners with content, paths, or scaffolds customized to their prior knowledge, needs, preferences, or contexts (Zhang et al., 2020). Personalization of AI's recommendation systems consists of three categories (Huang et al., 2023): content-based (i.e., grouping items according to content similarity), collaborative (i.e., providing recommendations based on similar attributes of learner groups), and hybrid recommendations (i.e., both content-based and collaborative). Personalization places

learners at the center of learning, given AI's adaptive capabilities that facilitate learners' interaction with the learning environment.

1.2.3 Learner control

Learner control originates from the Engagement Control Model (Sims, 1997), which focuses on the extent to which learners are making learning-related or navigational decisions while using technology. As an essential part of perceived interactivity, learner control over interaction reflects self-efficacy or confidence in performing activities with technology (Petersen et al., 2022). Learner control means allowing learners some control over AI with voluntary and instrumental actions that directly influence learning experiences (Liu & Shrum, 2002). In contrast to control by AI, learners may control learning content, sequence, or progress (Berendt et al., 2020). When learners perceive active control, their interactivity tends to be high because they are constantly controlling their experiences in interaction (Petersen et al., 2022).

1.2.4 Learning engagement

Learning engagement is another centered dimension of the Engagement Control Model (Sims, 1997), which emphasizes learners' involvement in interaction in a way designed to facilitate learning. Learning engagement supports deep cognitive processing in interaction. Learning engagement refers to the perception of learners that results from interacting with AI in learning and that generates involvement with the content studied (Shao & Chen, 2021). Engaged learners have high levels of involvement that prepare them better for active learning. Therefore, learners perceive more interactivity when they feel involved in learning experiences and engage in appropriate cognitive processes (Shin et al., 2016).

1.3 Potential associations of perceived interactivity of learner-AI interaction with personal factors and AI usage factors

1.3.1 Personal factors

Previous research has shown that the four personal factors might be associated with perceived interactivity of learner-AI interaction: ages, genders, education levels, and fields of education. As for ages, younger age groups tended to have a higher level of perceived interactivity with highly interactive technologies compared to older age groups (Fahmy & Ghoneim, 2019), since younger "digital natives" were likely to be more satisfied with the interactive digital products and older "digital immigrants" were generally more satisfied with static digital products (Kirk et al., 2015). In terms of genders, research indicated that females exhibited more positive perceptions of interactivity with technology than males (Shao & Chen, 2021). Females were more likely to exhibit interactive behaviors as more social and process-oriented, and males tended to display independent behaviors as more pragmatic and task-oriented (Lin et al., 2017). Regarding education levels, students at lower education levels perceived more interactivity with technology compared to students at higher education levels

(Fahmy & Ghoneim, 2019). As to fields of education, learners from scientific majors experienced higher levels of perceived interactivity with highly interactive technologies than learners from non-scientific majors (Fahmy & Ghoneim, 2019). However, since these findings were not revealed in the context of AI-supported learning environments, the relationship between the four personal factors and perceived interactivity of learner-AI interaction is far from conclusive.

1.3.2 AI usage factors

Besides the four personal factors, four other AI usage factors might be related to perceived interactivity of learner-AI interaction: AI tools, average duration of using AI each time, months of using AI in learning, and behavioral intentions to use AI. As AI technologies advance, AI tools are building stronger interactive capabilities to enhance the interactive experience for users according to user habits and preferences (Yin et al., 2021). Accordingly, different AI tools may exhibit different degrees of interactivity, and learners may have different levels of perceived interactivity depending on the tools used (Patwardhan & Murthy, 2015). Among AI tools used in education, generative AI increases perceived interactivity in digital learning environments to foster positive learning experiences (Chiu, 2023). Intelligent tutoring systems may also enhance perceived interactivity, which in turn improves learners' motivation, cognitive skills, and engagement (Ahuja et al., 2022). Various AI tools with interactive capabilities are also likely to improve learners' perceived interactivity with AI (Wang et al., 2023a). Furthermore, the duration that learners interact with AI may influence learners' perceived interactivity of learner-AI interaction since learners have various experiences in navigating AI resources, finding what they are looking for, and creating content that is understandable to them (Chou et al., 2022). Therefore, learners' average duration of using AI each time and months of using AI in learning should be taken into account. Additionally, learners' behavioral intentions to use AI are associated with their perceptions of the interaction process (Chou et al., 2022). While these studies analyzed the four AI usage factors that are potentially related to perceived interactivity of learner-AI interaction, they have not yet quantified the potential associations and confirmed the effects.

Therefore, to examine the potential associations, after developing and presenting the Perceived Interactivity of Learner-AI Interaction Scale, we assessed its relationship with the aforementioned personal and AI usage factors. The findings may shed light on further understanding of this topic.

1.4 Existing survey instruments and need for this study

Given the interdisciplinary nature of perceived interactivity, existing scales for measuring perceived interactivity have been developed and validated in fields other than education. For instance, scales on perceived interactivity were developed in business for mobile advertisements (e.g., Gao et al., 2010) and for websites (e.g., Liu, 2003; Wu, 2006), in computer science for usability in interaction with chatbots (e.g., Borsci et al., 2021), and in communication and media for web interaction (e.g., Sohn & Lee,

2005). However, in education, no scale has been found to measure perceived interactivity of learner-AI interaction.

As perceived interactivity of learner-AI interaction increasingly receives attention in educational research, the current instrument of measuring perceived interactivity of learner-AI interaction has relied mainly on a three-item instrument developed by Pituch and Lee (2006) to explore system interactivity of a web-based learning system in relevant studies (e.g., Girish et al., 2022; Wang et al., 2023a, b). Moreover, some studies (e.g., Sun & Hsu, 2013) drew on several items developed by other studies (e.g., Kettanurak et al., 2001) to measure separate dimensions. While use of several items has built a solid foundation for further research, a well-developed multi-item scale including all potential dimensions will provide advantages of accuracy, validity, and reliability (DeVellis & Thorpe, 2021). Therefore, there is a need for a comprehensive scale whose validity and reliability can be verified for measuring perceived interactivity of learner-AI interaction.

Furthermore, there is another need to use clearly and consistently defined concepts of interactivity and perceived interactivity in educational research (Petersen et al., 2022). Interactivity and perceived interactivity are sometimes used interchangeably. For example, interactivity refers to observable system functions in some studies (e.g., Alam & Mohanty, 2023; Sun & Hsu, 2013), but in other studies, interactivity means learners' subjective perceptions of their participation in interaction (Girish et al., 2022; Wang et al., 2023a). The inconsistent usage of the definitions of interactivity and perceived interactivity has complicated the comparability of empirical studies in the burgeoning field (Petersen et al., 2022; Quiring & Schweiger, 2008).

To summarize, a comprehensive, valid, and reliable scale for measuring perceived interactivity of learner-AI interaction has yet to be developed and validated, based on clear definitions of interactivity and perceived interactivity and consistent usage of the definitions in AI-supported learning environments. To address these needs, this study aims to develop and validate the Perceived Interactivity of Learner-AI Interaction Scale, in line with the definitions of interactivity and perceived interactivity presented in the section of the literature review. Additionally, this study tested and applied the scale to examine the relationship between perceived interactivity of learner-AI interaction and eight factors suggested in the literature: four personal factors (i.e., ages, genders, education levels, and fields of education) and four AI usage factors (i.e., AI tools, average duration of using AI each time, months of using AI in learning, and behavioral intentions to use AI in learning).

2 Method

This study aims to develop a measurement tool to examine learners' perceived interactivity of learner-AI interaction. To this end, a scale development method is appropriate because a scale is to measure phenomena that are believed to exist in theory but that cannot be assessed directly (DeVellis & Thorpe, 2021). We followed the procedure suggested by Churchill (1979) for developing and validating a scale (Yim et al., 2023): specifying the theoretical dimensions of the concept in literature review, generating an item pool, collecting data, purifying measure by the exploratory factor

analysis (EFA) and the confirmatory factor analysis (CFA), and refining and validating the scale. In the process, the EFA aims to identify the underlying factor structure in the dataset that aligns with theoretical dimensions, thereby facilitating the understanding of the construct, and the CFA tries to test the factor structure during the EFA in terms of whether it can be confirmed or not (Hair et al., 2010). Since the first step of specifying the dimensions of the concept has been completed in the above literature review, the remaining steps are implemented below.

2.1 Generation of an item pool

An item pool was generated at four stages. First, the relevant literature was reviewed, and the scales measuring similar properties were investigated to generate an initial item pool. In particular, the Engagement Control Model (Sims, 1997) was employed to identify the theoretical infrastructure. Based on this model, we carefully examined other scholars' studies on perceived interactivity and selected items in education. The literature review produced an initial item pool of 20 items, measuring responsiveness (Gao et al., 2010; Liu, 2003), learner control (Liu, 2003; Wu, 2006), learning engagement (Fredricks et al., 2004; Gallini & Moely, 2003; McMillan & Hwang, 2002), and personalization (Alimamy & Gnoth, 2022; Xu et al., 2011). The item pool was measured on a 6-point Likert-type scale with the options: "strongly disagree," "disagree," "somewhat disagree," "somewhat agree," "agree," and "strongly agree." The even-numbered 6-point scale was adopted because an even number tends to force respondents to choose a definite option, potentially reducing the likelihood of selecting a neutral option without careful consideration (Brown, 2000) and an expression of "don't know" (Raaijmakers et al., 2000). Moreover, the 6-point Likert scale has the advantages of minimizing risks associated with deviations in personal decision-making and enhancing discrimination and reliability of responses in self-reported surveys (Chomeya, 2010).

Second, we interviewed 16 students who had the experience of using AI in learning about their thoughts on perceived interactivity of learner-AI interaction. The transcripts of the interviews were analyzed, and the main content was added to the item pool appropriately. These students were from four universities, each located in one of the four cities in China: Beijing, Shanghai, Shenzhen, and Guangzhou. The four cities were also where this study collected surveys. The first criterion for determining the four cities is that these cities have sustained advantages in AI development and application in China, as suggested by their top rankings in "China AI City Competitiveness," including AI infrastructure, products, developers, users, etc. (TMT, 2024). The widespread development and adoption of AI in the cities may give learners broad access to diverse AI tools. Another criterion is that, in these cities, AI education was initiated relatively early (Huang, 2021), and AI has been broadly integrated into education (Knox, 2020). This offers a practical foundation for learners from different backgrounds to engage with AI across various educational settings. Together, the criteria for selecting the four cities were designed to access a broad range of university students with extensive experience in using AI in education.

The 16 students in the interview included eight undergraduate students, five master's students, and three doctoral students who majored in economics, psychology,

law, linguistics, visual arts, physics, electrical engineering, and clinical medicine. Their mean age was 22.1, including seven females and nine males. The interviews were analyzed by three researchers. Considering that the interviews captured items that were not covered in the initial item pool, we added three additional items, resulting in a total of 23 items.

Third, concerning the content validity, we sent the 23 items to seven experts to obtain their feedback on whether the items were suitable for evaluating the anticipated dimensions and whether the expressions of the items were clear and accurate. All the experts possessed doctoral degrees. Three of them had publications on AI in education, two on education measurement and evaluation, and two on linguistics. In addition, the experts were provided with a form that introduced the purpose of the scale, target participants, items, and anticipated dimensions. They were asked to evaluate each item with the options: "suitable," "requiring improvement," and "unsuitable." Their feedback suggested the removal of six items and the improvement of two items. We made improvements to the two items in line with the experts' feedback, obtaining an item pool of 17 items.

Fourth, a preliminary test of the scale was conducted with 20 students among a sample of our target participants. Participants shared opinions on whether the items were clear and appropriate in relation to their learning experience. Their feedback indicated no need to revise the items. The final 17-item scale included five items on responsiveness, three on learner control, six on learning engagement, and three on personalization.

2.2 Data collection

Data were collected in China via a widely used professional online survey platform, Wenjuanxing, in January 2024. This survey started with detailing the purpose and protocol of the study. University students who had the experience of using AI in learning were invited to participate. Participants provided informed consent before answering the questions, and they could withdraw at any point.

This study involved two distribution phases for data collection. Initially, 600 surveys were collected for the EFA phase to identify the factor structure of the 17 items. Subsequently, 400 surveys were collected for the CFA phase to test and confirm the factor structure of the items derived from the EFA. In the total of 1000 surveys, there was no missing value, given that the surveys could be submitted only after all the questions were answered on the platform. Filters were used to exclude data via three criteria: (1) if the total length of using AI in learning was less than two months, (2) if the respondents were not students from universities of the four cities in China: Beijing, Shanghai, Shenzhen, and Guangzhou, and (3) if the surveys were completed within 90 s. After applying the three exclusion criteria, this study obtained 422 participants for the EFA dataset and 306 participants for the CFA dataset. In scale development studies, it is recommended to determine the proper sample size by multiplying the number of items by ten or twenty, and studies may employ a parsimonious and adequate sample size, guided by conceptual and practical considerations (Hair et al., 2010). When evaluated together, it is found that the sample sizes for both the EFA and the CFA are sufficient for further analyses.

For the EFA dataset, 56.9% were male, 66.1% were aged between 17 and 20 years, and 96% were undergraduate students. For the CFA dataset, 60.1% were male, 66% were between 17 and 20 years old, and 93.8% were undergraduate students. Regarding the AI tool most frequently used in learning, participants filled in the survey according to their personal experiences. Based on the taxonomies of student-focused AI tools in education by Holmes and Tuomi (2022) and by Zhang and Aslan (2021), we modified and classified the tool into five categories: generative AI (e.g. ChatGPT and Midjourney), intelligent tutoring system (e.g. Spark and LAIX), automatic formative assessment (e.g., iFLYTEK scoring and Zuoyebang), personalized learning system (e.g., course-specific expert system and self-developed system for teaching and learning), and writing assistant (e.g., Grammarly and Baidu Translate). For the EFA dataset, the most frequently used AI tool in learning was generative AI (37%), followed by personalized learning system (19.2%), intelligent tutoring system (17.5%), automatic formative assessment (13.5%), and writing assistant (12.8%). For the CFA dataset, the predominant AI tool used in learning was also generative AI (40.8%), with personalized learning system (19%), intelligent tutoring system (14.1%), automatic formative assessment (13.7%), and writing assistant (12.4%) in usage. It is noteworthy that participants were reminded during the survey to specifically refer to their interaction with the AI tools they used most frequently, which ensured that their subsequent responses were directly related to these tools mentioned. Detailed demographic information of the two datasets is presented in Table 2.

2.3 Factor structure identification by the EFA and CFA

In order to identify the factor structure of the Perceived Interactivity of Learner-AI Interaction Scale, both the EFA and CFA were conducted with Mplus Version 8.3. The EFA was performed to reveal the factor structure of the scale and identify which items were clustered within a common factor. The purpose of the analysis is to provide construct validity of the scale and to find a small number of conceptually meaningful variables (i.e., factors or dimensions) by gathering interrelated variables (Field, 2018). In the EFA, Maximum Likelihood Robust was employed as the parameter estimator to adjust for the asymmetric distribution of a few items in the dataset with skewness higher than |1| and kurtosis higher than |2| (Noar, 2003). We used the oblique GEOMIN rotation allowing for correlations between factors (Muthén & Muthén, 2017). When determining the value of factor loading (i.e., the correlation of an item with its associated factor), considering that the factor loading of 0.4 and above is considered substantial (Floyd & Widaman, 1995), we retained the items with the factor loading of higher than 0.4 on one factor (Hair et al., 2010). We assessed the EFA model fit by comparing multiple model fit indices (Anderson & Gerbing, 1988). The suggested model is "good" when the fit values of the model fit indices are within the range: $0.00 < \chi 2/df < 3.00$, $0.00 <$ RMSEA (i.e., the root mean square error of approximation) < 0.05, $0.00 <$ SRMR (i.e., the standardized root mean square residual) < 0.05, $0.95 <$ TLI (i.e., the Tucker-Lewis Index, also known as the non-normed fit index: NNFI) < 1.00, $0.95 <$ CFI (i.e., the comparative fit index) < 1.00, and "acceptable" within the range: $3.00 < \chi 2/df < 5.00$, $0.05 <$ RMSEA < 0.10, $0.05 <$ SRMR < 0.10, $0.90 <$ TLI < 0.95, $0.90 <$ CFI < 0.95 (Yurdakul et al., 2012).

Table 2 Demographic information for the EFA and CFA datasets

Characteristic	Category	EFA (n=422)		CFA (n=306)	
		Frequency	Percentage	Frequency	Percentage
Gender	Male	240	56.9%	184	60.1%
	Female	182	43.1%	122	39.9%
Age	17–20	279	66.1%	202	66.0%
	21–25	132	31.3%	97	31.7%
	Above 26	11	2.6%	7	2.3%
Education level	Bachelor's degree	405	96.0%	287	93.8%
	Master's degree	14	3.3%	15	4.9%
	Doctor's degree	3	0.7%	4	1.3%
Field of education	Arts and humanities	44	10.4%	29	9.5%
	Social sciences	164	38.9%	112	36.6%
	Science	63	14.9%	51	16.7%
	Engineering	133	31.5%	102	33.3%
	Medicine	18	4.3%	12	3.9%
City	Beijing	116	27.5%	91	29.7%
	Shanghai	162	38.4%	118	38.6%
	Shenzhen	86	20.4%	51	16.7%
	Guangzhou	58	13.7%	46	15.0%
AI tool	Generative AI	156	37.0%	125	40.8%
	Intelligent tutoring system	74	17.5%	43	14.1%
	Automatic formative assessment	57	13.5%	42	13.7%
	Personalized learning system	81	19.2%	58	19.0%
	Writing assistant	54	12.8%	38	12.4%
Year/month of using AI in learning	Between 2 months and 1 year	150	35.5%	103	33.7%
	Between 1 and 2 years	142	33.6%	100	32.7%
	Between 2 and 3 years	61	14.5%	54	17.6%
	Between 3 and 7 years	53	12.6%	41	13.4%
	Between 7 and 12 years	16	3.8%	8	2.6%
Average duration of using AI in learning each time	Less than 1 h	343	81.3%	256	83.7%
	Between 1 h and 2 h	34	8.1%	22	7.2%
	More than 2 h	45	10.7%	28	9.2%

Note EFA=exploratory factor analysis. CFA=confirmatory factor analysis

The decision regarding the number of factors was based on the lower limit of the eigenvalue as 1 and a parallel analysis. A parallel analysis was conducted to provide supplementary criteria in determining the number of factors extracted. One thousand random data matrices were generated, matching the cases and variables in the original dataset. Eigenvalues generated from the original dataset were compared with both the average and 95th percentile eigenvalues generated from the random data matrices (Hayton et al., 2004). Factors were retained if the eigenvalues of the factors from the original data were higher than both the average and 95th percentile eigenvalues from the random data matrices.

After the EFA, we performed a CFA with the second dataset to examine whether the factor structure suggested by the EFA was confirmed. Again, Maximum Likelihood Robust was performed to improve estimates for a few asymmetric items. The

CFA model fit was also assessed based on the aforementioned model fit indices: $\chi 2/df$, RMSEA, SRMR, TLI, and CFI.

2.4 Scale refinement and validation

In the scale refinement, validity and reliability of the scale were assessed with the SPSS Version 26. The convergent and discriminant validity of the scale was assessed by examining composite reliability (CR) and average variance extracted (AVE) values of the scale (Hair et al., 2010). For convergent validity, the CR should be above 0.7, and the AVE should be above 0.5. For discriminant validity, the square root of each factor's AVE should be greater than the factor's correlation with other factors in the model and should be above 0.5 (Fornell & Larcker, 1981). To evaluate the reliability of the scale, internal consistency tests include Cronbach's alpha reliability tests. A reliability coefficient higher than 0.70 is considered "good", and when it is close to 1.00, it is considered "very good" (Fraenkel et al., 2012).

After these, utility tests of the scale were performed to examine the relationship between the scale and the eight related variables. First, to compare mean perceived interactivity scores within the five documented categorical variables: genders, education levels, fields of education, AI tools, and average duration of using AI each time, independent samples t-tests and ANOVAs were conducted. Second, to examine the relationship between mean perceived interactivity scores and the three documented ordinal and continuous variables: ages, months of using AI in learning, and behavioral intentions to use AI in learning, a correlation analysis was performed. Behavioral intentions to use AI in learning were measured by the adapted behavioral intention measures of using AI in learning by Liu and Ma (2023) and of using educational technology by Sun and Mei (2022). The six adapted items are presented in Table 3. Third, the strength of these relationships was assessed together with a multiple regression analysis. Results of the regression analysis reveal criterion validity of the scale, indicating the potential of using the scale as a predictive tool and providing insights into the discussion.

Table 3 *Measure of behavioral intentions to use AI in learning (Cronbach's alpha = 0.942, N = 728)*

Item	Mean	Standard deviation	Alpha if deleted
1. I am willing to spend time and effort to learn how to use AI better.	4.63	1.051	0.941
2. I expect to use AI frequently in the future for learning.	4.69	1.041	0.931
3. I plan to use AI for my learning needs in the future.	4.71	1.000	0.927
4. I would like to use AI to support my learning in the future.	4.73	1.021	0.925
5. I think I will utilize AI for my learning.	4.74	0.972	0.928
6. I will recommend using AI for learning to other learners.	4.74	1.034	0.932

Note Measured on a 6-point Likert scale. 1 refers to strongly disagree and 6 refers to strongly agree

Table 4 EFA goodness of fit indices

EFA-extracted models	χ^2	*df*	χ^2/df	RMSEA	SRMR	TLI	CFI
1-Factor	1063.813***	119	8.940	0.137	0.098	0.588	0.639
2-Factor	671.584***	103	6.520	0.114	0.074	0.713	0.783
3-Factor	382.630***	88	4.348	0.089	0.049	0.826	0.887
4-Factor	151.590***	74	2.049	0.050	0.021	0.946	0.970

Note χ^2 = Chi-square. *df*=degree of freedom. RMSEA=root mean square error of approximation. SRMR=standardized root mean square residual. TLI=Tucker-Lewis Index. CFI=comparative fit index

*** $p<.001$

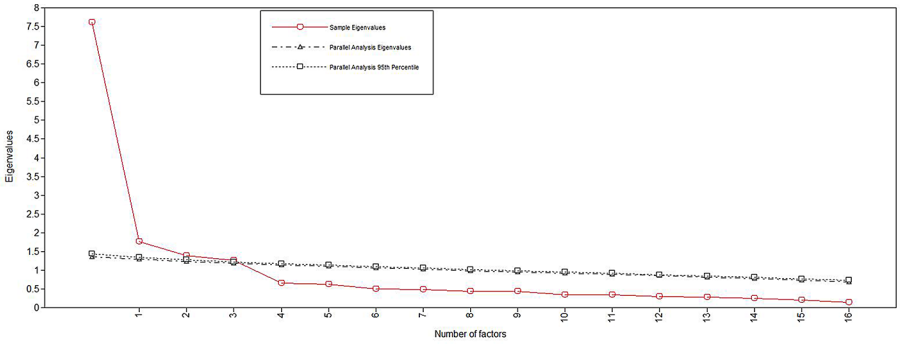


Fig. 2 Parallel analysis. *Note* The red line indicates the eigenvalues generated from the original dataset. Two other lines indicate the average eigenvalues and 95th percentile eigenvalues generated from the parallel random data

3 Results

3.1 Results of the EFA

The EFA results supported a four-factor model. As Table 4 shows, all the model fit indices consistently showed that when the number of factors extracted increased from one to four, the model fit improved. One-factor and two-factor models had poor fits based on $\chi^2/df>5$, RMSEA>0.1, TLI<0.9, CFI<0.9, despite their SRMR<0.1. The three-factor model showed a better fit for its SRMR was good (SRMR<0.05) and χ^2/df and RMSEA were acceptable ($\chi^2/df<5$, RMSEA<0.1), but its TLI and CFI were poor (TLI<0.9, CFI<0.9). However, the four-factor model showed the best fit for its χ^2/df, SRMR, and CFI were good ($\chi^2/df<3$, SRMR<0.05, CFI>0.95) and its RMSEA and TLI were acceptable (RMSEA>0.5, TLI>0.9).

The parallel analysis also showed a similar pattern, indicating that four factors should be extracted. As displayed in Fig. 2, the first four factors had eigenvalues higher than the average and 95th percentile eigenvalues generated from the parallel random data. To be specific, all the first four factors produced actual eigenvalues higher than one (Factor 1=7.607, Factor 2=1.765, Factor 3=1.387, Factor 4=1.263), which were greater than their respective average and 95th percentile eigenvalues, as shown in Table 5. Therefore, we decided that a four-factor model should be retained.

Table 5 Eigenvalues of parallel analysis

Factor	Actual eigenvalues	Average eigenvalues	95th percentile eigenvalues
Factor 1	7.607	1.363	1.430
Factor 2	1.765	1.290	1.342
Factor 3	1.387	1.233	1.278
Factor 4	1.263	1.184	1.223
Factor 5	0.649	1.141	1.176
Factor 6	0.617	1.100	1.132

Table 6 EFA results: Mean, standard deviation, skewness, kurtosis, and factor loadings

Factor	Item	Mean	*SD*	Skewness	Kurtosis	Factor loading of each item			
						Factor 1	Factor 2	Factor 3	Factor 4
Responsiveness	Item 1	4.773	1.204	~1.370	2.628	0.950			
	Item 2	4.720	1.268	~1.129	1.721	0.891			
	Item 3	4.557	1.280	~0.789	0.807	0.751			
	Item 4	4.149	1.682	~0.396	~0.304	0.505			
	Item 5	4.552	1.385	~0.858	0.791	0.504			
Learner control	Item 6	4.211	1.493	~0.400	~0.264		0.707		
	Item 7	3.877	1.786	~0.257	~0.539		0.901		
	Item 8	4.197	1.362	~0.360	~0.195		0.744		
Learning engagement	Item 9	4.377	1.168	~0.441	0.267			0.461	
	Item 10	4.472	1.221	~0.777	0.801			0.667	
	Item 11	4.545	1.073	~0.748	1.239			0.483	
	Item 12	4.270	1.235	~0.516	0.257			0.713	
	Item 13	4.405	1.118	~0.514	0.483			0.658	
	Item 14	4.543	1.049	~0.717	1.000			0.777	
Personalization	Item 15	4.235	1.516	~0.712	0.333				0.919
	Item 16	4.382	1.388	~0.668	0.236				0.418
	Item 17	4.277	1.622	~0.674	~0.055				0.730

Note SD = standard deviation

Table 6 presents GEOMIN rotated loadings for the four-factor model. The factor loadings of the 17 items varied between 0.504 and 0.950 for the first factor, responsiveness; between 0.707 and 0.901 for the second factor, learner control; between 0.461 and 0.777 for the third factor, learning engagement; and between 0.418 and 0.919 for the fourth factor, personalization. All the factor loadings exceeded 0.4, indicating substantial values, so the four factors of the scale were identified.

3.2 Results of the CFA

The CFA was conducted to confirm the four-factor structure with 17 items identified by the EFA. As a result of the CFA with the Maximum Likelihood Robust estimator, the model fit values were found to be χ^2/df (χ^2 = 242.734, df = 113) = 2.148, $p < .0001$, RMSEA = 0.061, SRMR = 0.062, TLI = 0.926, and CFI = 0.938. According to these values, it can be observed that $\chi 2/df$ indicates a "good" model fit, and RMSEA,

SRMR, TLI, and CFI indicate an "acceptable" model fit. In other words, this model fit indicates that the four factors of the scale can be confirmed by the CFA, so we did not make any modifications to the model or remove any items.

The path diagram with factor loads and error variances of the items is displayed in Fig. 3. The factor loads of the scale varied between 0.704 and 0.878 for the first factor (i.e., responsiveness), between 0.814 and 0.832 for the second factor (i.e., learner control), between 0.748 and 0.847 for the third factor (i.e., learning engagement),

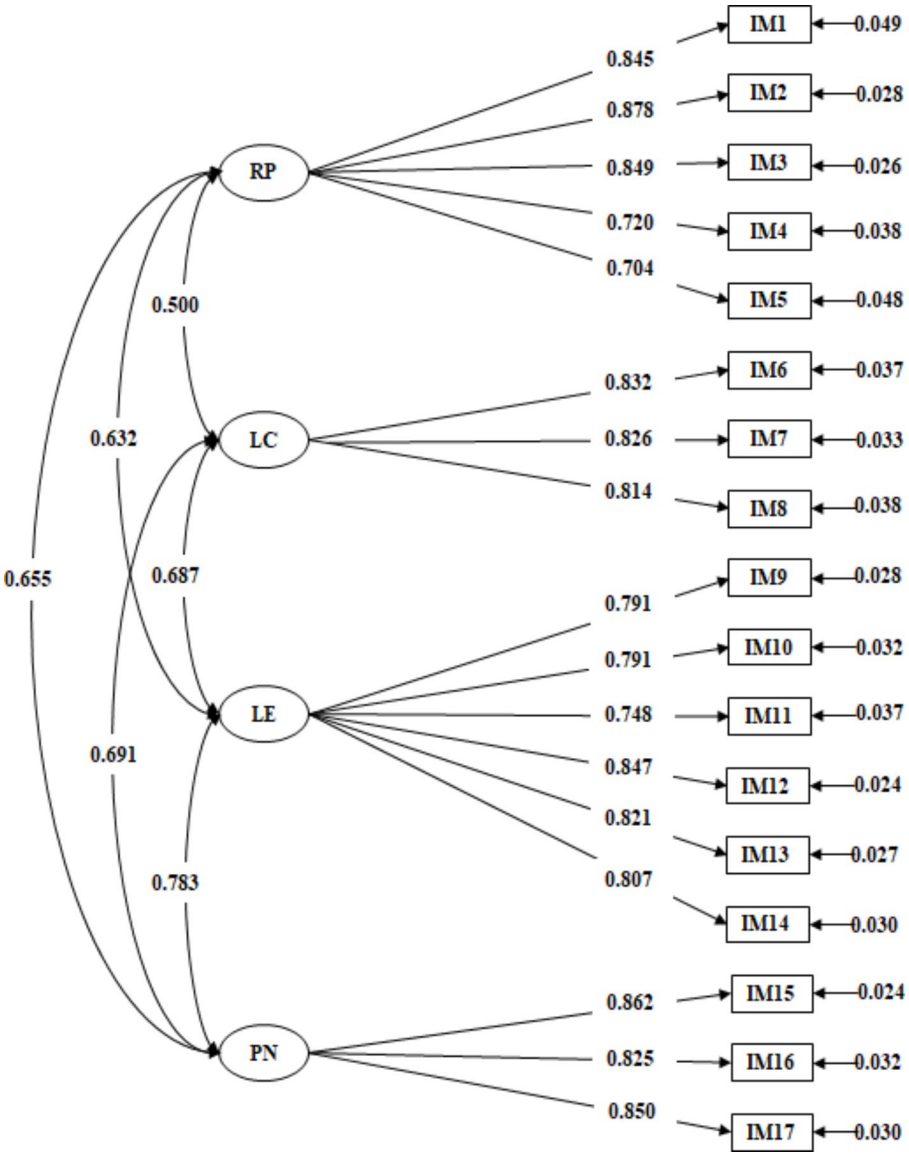


Fig. 3 The path diagram of the CFA. *Note* RP = responsiveness. LC = learner control. LE = learning engagement. PN = personalization

and between 0.825 and 0.862 for the fourth factor (i.e., personalization). All the coefficients are significant at $p < .0001$. These indicate that the factor loadings are sufficient, and the items show high correlations (Cohen & Holliday, 1982).

3.3 Convergent validity, discriminant validity, and reliability

After the CFA confirms the factor structure suggested by the EFA, it is recommended to examine composite reliability (CR) and average variance extracted (AVE) values of the scale for its convergent validity (Fornell & Larcker, 1981; Hair et al., 2010). The CR should be above 0.7, and the AVE should be above 0.5. As shown in Table 7, the CR values of the four factors (i.e., 0.900, 0.864, 0.915, and 0.883 respectively) are all above 0.7. The AVE values (i.e., 0.644, 0.679, 0.642, and 0.715 respectively) are all above 0.5. Thus, the convergent validity for the scale is adequate.

Cronbach's alpha values were calculated to examine the consistency of the items in the scale. Results in Table 7 showed that the Cronbach's alpha values of the four factors were 0.895 for responsiveness, 0.820 for learner control, 0.915 for learning engagement, and 0.868 for personalization. The Cronbach's alpha value of the whole scale was 0.948. A Cronbach's alpha value is considered "good" when it is above 0.70 and "very good" when it is close to 1.00 (Fraenkel et al., 2012). Therefore, the reliability of this scale can be interpreted as "very good".

Regarding discriminant validity, according to Fornell and Larcker (1981), the square root of each factor's AVE should be higher than the factor's correlation with other factors in the model and should be above 0.5. Table 8 shows that the square

Table 7 CFA results: Mean, standard deviation, skewness, kurtosis, convergent validity, and divergent validity of factors

Factor	Item	Mean	*SD*	Skewness	Kurtosis	CA	CR	AVE
Responsiveness	Item 1	4.745	1.223	~1.153	1.855	0.895	0.900	0.644
	Item 2	4.784	1.019	~0.894	1.444			
	Item 3	4.621	1.137	~0.640	0.520			
	Item 4	4.369	1.403	~0.330	~0.386			
	Item 5	4.667	1.196	~0.676	0.279			
Learner control	Item 6	4.206	1.347	~0.243	~0.370	0.820	0.864	0.679
	Item 7	4.016	1.755	~0.317	~0.428			
	Item 8	4.268	1.222	~0.209	~0.213			
Learning engagement	Item 9	4.314	1.215	~0.423	0.082	0.915	0.915	0.642
	Item 10	4.454	1.111	~0.449	0.172			
	Item 11	4.520	0.942	~0.291	0.013			
	Item 12	4.314	1.215	~0.423	0.242			
	Item 13	4.458	1.124	~0.349	0.021			
	Item 14	4.526	1.066	~0.506	0.485			
Personalization	Item 15	4.431	1.173	~0.533	0.451	0.868	0.883	0.715
	Item 16	4.542	1.078	~0.455	0.117			
	Item 17	4.503	1.230	~0.468	~0.039			

Note *SD* = standard deviation. CA = Cronbach's alpha. CR = composite reliability. AVE = average variance extracted

Table 8 Correlations among factors and discriminant validity

Factor	Factor 1	Factor 2	Factor 3	Factor 4
1. Responsiveness	**0.802**			
2. Learner control	0.500***	**0.824**		
3. Learning engagement	0.632***	0.687***	**0.801**	
4. Personalization	0.655***	0.691***	0.783***	**0.846**

Note Diagonal values in bold represent the square root of the average variance extracted (AVE) of the factors

*** $p < .001$

Table 9 Comparison of mean perceived interactivity scores on categorical variables

Characteristic	Category	*N*	Mean	*SD*	*t/F*	*p*
Gender	Male	424	75.375	14.152	~0.380	.743
	Female	304	75.697	12.274		
Education level	Bachelor's degree	692	75.408	13.447	0.453	.636
	Master's degree	29	77.138	12.415		
	Doctor's degree	7	78.857	12.628		
Field of education	Arts and humanities	73	76.644	13.398	0.344	.848
	Social sciences	276	75.337	13.739		
	Sciences	114	76.351	14.461		
	Engineering	235	75.021	12.555		
	Medicine	30	74.967	12.837		
AI tool	Generative AI	281	73.655	11.061	58.440	.000
	Intelligent tutoring system	117	84.752	11.128		
	Automatic formative assessment	99	66.465	11.284		
	Personalized learning system	139	82.899	11.800		
	Writing assistant	92	67.989	14.146		
Average duration of using AI each time	Less than 1 h	599	74.738	13.231	5.714	.003
	Between 1 h and 2 h	56	79.411	12.942		
	More than 2 h	73	78.849	14.168		

Note *N* = the number of participants. *SD* = standard deviation

roots of the AVE of all the factors are higher than the inter-factor correlations. Therefore, it has been observed that each factor has a high discriminant power in the scale.

3.4 Utility of the scale

To test and apply the scale, independent samples *t*-tests and ANOVAs were conducted to compare mean perceived interactivity scores on related characteristics. Independent samples *t*-tests compared the mean difference between males and females. As shown in Table 9, there was no significant difference between males and females (t = -0.380, $p = .743$). ANOVAs compared mean perceived interactivity scores on education levels, fields of education, AI tools, and average duration of using AI each time. There was no significant difference in education levels, F (2, 725) = 0.453, $p = .636$, and in fields of education, F (4, 723) = 0.344, $p = .848$.

As for AI tools, results showed that there was a significant difference, F (4, 723) = 58.440, p < .001. Pos hoc comparisons using Tukey's Honestly Significance Difference found that the mean perceived interactivity score was significantly different between "generative AI" and "writing assistant" (p = .001, 95% C.I. = [1.831, 9.500]); between "generative AI" and "intelligent tutoring system" (p < .001, 95% C.I. = [-14.610, -7.585]); between "generative AI" and "automatic formative assessment" (p < .001, 95% C.I. = [3.459, 10.921]); between "generative AI" and "personalized learning system" (p < .001, 95% C.I. = [-12.555, -5.934]); between "writing assistant" and "intelligent tutoring system" (p < .001, 95% C.I. = [-21.211, -12.315]); between "writing assistant" and "personalized learning system" (p < .001, 95% C.I. = [-19.201, -10.620]); between "intelligent tutoring system" and "automatic formative assessment" (p < .001, 95% C.I. = [13.928, 22.647]); and between "personalized learning system" and "automatic formative assessment" (p < .001, 95% C.I. = [12.236, 20.633]). There was no significant difference between "writing assistant" and "automatic formative assessment" (p = .896); and between "intelligent tutoring system" and "personalized learning system" (p = .713).

Regarding the average duration of using AI each time, results indicated a significant difference, F (2, 725) = 5.714, p = .003. Pos hoc comparisons using Tukey's Honestly Significance Difference indicated that the mean perceived interactivity score for "between one and two hours" was significantly higher than "less than one hour" (p = .033, 95% C.I. = [0.306, 9.040]). The mean perceived interactivity score for "more than two hours" was also significantly higher than "less than one hour" (p = .034, 95% C.I. = [0.238, 7.985]). No significant difference was found between "between one and two hours" and "more than two hours" (p = .969).

Relationships between mean perceived interactivity scores and ordinal and continuous variables were investigated with a correlation analysis, as displayed in Table 10. Results showed significant relationships between perceived interactivity and the following variables: age, month of using AI in learning, and behavioral intention to use ($r \geq .101$, $p < .01$).

Given these results, a multiple linear regression examined the following model equation: perceived interactivity = age + month of using AI in learning + behavioral intention to use. This model was significant, F (3, 724) = 435.904, p < .001, and explained 64.4% of the variance (i.e., $R^2 = 0.644$). As shown in Table 11, month of using AI in learning was a significant positive predictor ($\beta = 0.048$, $t = 2.138$, $p = .033$),

Table 10 Correlation analysis of ordinal and continuous variables

Variable	Mean	*SD*	Correlation			
			Age	Month of using AI in learning	Behavioral intention to use	Perceived interactivity
Age	20.270	2.134	1			
Month of using AI in learning	23.090	23.964	0.202**	1		
Behavioral intention to use	28.254	5.385	0.091*	0.061	1	
Perceived interactivity	75.510	13.392	0.104**	0.101**	0.800**	1

Note N = 728. *SD* = standard deviation

** p < .01. * p < .05

Table 11 Regression analysis of perceived interactivity of learner-AI interaction

Independent variables	*B*	*SE B*	*β*	*t*	*p*
Age	0.134	0.143	0.021	0.937	.349
Month of using AI in learning	0.027	0.013	0.048	2.138	.033
Behavioral intention to use	1.978	0.055	0.795	35.670	.000

Note B=the unstandardized coefficient. *SE* B=the standard error of the unstandardized coefficient. β=the standardized coefficient

indicating that as months spent using AI in learning increased, so did perceived interactivity scores. Age was not a significant positive predictor ($\beta=0.021$, $t=0.937$, $p=.349$). Behavioral intention to use was the most influential predictor ($\beta=0.795$, $t=35.670$, $p<.001$) and provided solid evidence for the criterion validity of the scale.

4 Discussion

Behavioral research investigating the influence of AI on learners' behavioral development and on their learning of new skills and competences is increasing rapidly (Benvenuti et al., 2023; Chai et al., 2021). A growing body of research indicates that learners who perceive higher levels of interactivity in learner-AI interaction are not only more likely to participate actively in intelligent teaching behaviors (Wang et al., 2023a) and become engaged in the learning process (Zhang et al., 2021), but also exhibit relationships between perceived interactivity and other related aspects, such as motivation, perceived enjoyment, divergent thinking, creativity, academic performance, and well-being (Afanasyeva et al., 2024; Benvenuti et al., 2023; Wang & Cheung, 2024). Despite the importance, however, no scale for measuring perceived interactivity of learner-AI interaction has been developed in the field of education. Moreover, the inconsistent usage of the definitions of interactivity and perceived interactivity has complicated the comparability of empirical studies in the emerging field (Petersen et al., 2022). To address this noteworthy topic, we developed, validated, and presented such a scale: the Perceived Interactivity of Learner-AI Interaction Scale.

4.1 Development and validation of the Perceived Interactivity of Learner-AI Interaction Scale

Based on a literature review, student interviews, and expert opinions, four key dimensions were first identified. The instrument was developed by assessing the content and construct validity of items in each dimension. The EFA on a development sample revealed the same four dimensions, which were subsequently validated through the CFA on another confirmatory sample. The final version of the scale, supported by evidence of reliability and validity, includes four dimensions with 17 items, as shown in Table 12. The "responsiveness" dimension covers items measuring learners' perceptions of AI's responses in terms of speed, appropriateness, and relevance. The "learner control" dimension includes items measuring learners' perceptions of their control over the learning content, pace, and experiences. The "learning engage-

Table 12 Final version of the Perceived Interactivity of Learner-AI Interaction Scale

Dimension	Item	Related literature
Responsiveness	1. AI processes my response quickly.	Gao et al., 2010; Liu, 2003
	2. AI is fast in responding to my query.	
	3. I can get instantaneous information when I respond to AI.	
	4. I can obtain the information I want from AI without any delay.	
	5. I can communicate with AI directly for further questions.	
Learner control	6. While using AI, I choose freely what I want to learn.	Liu, 2003; Wu, 2006
	7. While I am using AI, my actions decide the kind of learning experiences I get.	
	8. While using AI, I am in total control over the pace of my use of AI.	
Learning engagement	9. While using AI, I am satisfied with the quality of my learning experiences.	Fredricks et al., 2004; Gallini & Moely, 2003; McMillan & Hwang, 2002
	10. I become more interested in using AI for learning.	
	11. AI keeps my attention.	
	12. I am involved in the learning experiences using AI.	
	13. I actively use AI in learning.	
	14. I feel my learning intellectually stimulating while using AI.	
Personalization	15. AI satisfies my specific learning needs.	Alimamy & Gnoth, 2022; Xu et al., 2011
	16. AI takes my needs as its own preferences.	
	17. AI is personalized to my learning context.	

ment" dimension consists of items that elicit learners' responses for three types of engagement: emotional, behavioral, and cognitive engagement. The "personalization" dimension involves items that measure learners' perceptions of how well AI responds to their learning needs, preferences, and contexts.

This study provides a new instrument for the field of education, specifically developed to measure learners' perceived interactivity of learner-AI interaction using comprehensive dimensions grounded in the Engagement Control Model, which sets the instrument apart from other relevant studies from three aspects. First, it is an instrument developed to measure learners' perceived interactivity in learner-AI interaction in the field of education. Previous studies have developed people's perceived interactivity, but many were developed for other fields, such as computer science (e.g., Borsci et al., 2021) and business (e.g., Gao et al., 2010). Second, this instrument, grounded in clearly defined distinctions between interactivity and perceived interactivity from the literature, clarifies the currently inconsistent usage of these concepts and provides a basis for comparability in future empirical studies (Girish et al., 2022). Third, the instrument has been validated to include comprehensive dimensions grounded in the Engagement Control Model. Some studies have mainly used several items to evaluate separate dimensions of system interactivity (e.g., Sun & Hsu, 2013), but this instrument extends these studies by incorporating all potential dimensions related to the topic, as indicated in the Engagement Control Model.

4.2 Utility of the scale

To examine the utility of the scale and further understand perceived interactivity of learner-AI interaction, we tested and applied the scale across eight variables as indicated in literature: ages, genders, education levels, fields of education, AI tools, average duration of using AI each time, months of using AI in learning, and behavioral intentions to use. Results indicated that perceived interactivity of learner-AI interaction was associated with four of the variables: AI tools, average duration of using AI each time, months of using AI in learning, and behavioral intentions to use.

Specifically, regarding AI tools, perceived interactivity of learner-AI interaction scored the highest with intelligent tutoring systems, followed by personalized learning systems, generative AI, writing assistant, and automatic formative assessment. The results are in line with many previous studies. One reason why intelligent tutoring systems produced the highest score may be explained by the interaction granularity hypothesis, suggesting that the fine granularity (i.e., grain size) of intelligent tutoring systems reduces reasoning difficulties for learners in learning and enhances the scaffolding provided by the systems during learning (Liu et al., 2024). In other words, intelligent tutoring systems provide learners with many interaction opportunities during learning by providing detailed learning steps, substeps, and even answers (Du Boulay, 2016). Compared to intelligent tutoring systems, personalized learning systems (e.g., self-developed system for programming) and generative AI need more learner-initiated interaction such as phrasing meaningful questions (Gao et al., 2024; Tang et al., 2020), which may influence the interactive process of personalized learning. Automatic formative assessment and writing assistant need less learner-initiated interaction for directly providing contextualized and generic feedback on specific issues (Zhu et al., 2020). Another reason may be related to the ICAP (Interactive, Constructive, Active, and Passive) framework (Chi, 2009; Gao et al., 2024). This framework suggests that interactive activities are likely to be better than constructive activities, which in turn may outperform active activities, with passive activities being the last. The explanation predicts that learners working with more interactive systems (e.g., intelligent tutoring systems, personalized learning systems, and generative AI) are more likely to exhibit interactive behaviors, while those using less interactive systems (e.g., automatic formative assessment and writing assistant) may exhibit guessing and imitation, reflecting active or passive learning behaviors.

As for the duration of using AI in learning, both related variables, including average duration of using AI each time and months of using AI in learning, produced significant results. To be specific, perceived interactivity scores when lasting for "less than one hour" were found to be significantly lower than two other categories: "between one and two hours" and "more than two hours". Months of using AI in learning also produced significant results, suggesting that increased months of using AI in learning were associated with higher perceived interactivity scores. The results align with previous research to indicate that the longer duration of interacting with interactive technology is related to higher motivation and more perceived interactivity (Barut Tugtekin & Dursun, 2022). However, there might be boundary conditions that involve characteristics of learners and learning contexts (Liu & Shrum, 2002; Wang & Cheung, 2024). Given that the numbers of participants in the two categories

of "between one and two hours" and "more than two hours" are much lower than "less than one hour", the findings need more evidence from future research.

In terms of behavioral intentions to use, it was significantly associated with perceived interactivity of learner-AI intention. This indicates that learners who scored higher on the Perceived Interactivity of Learner-AI Interaction Scale showed stronger intentions to use AI in learning. The results align with the findings of existing research that learners interacting with virtual reality in active-interactivity learning environments have enriched learning experience and enhanced motivation, compared to those in passive-interactivity environments (Roussou & Slater, 2020), thereby showing stronger behavioral intentions to use (Barrett et al., 2021). Similarly, another study by Kim and Kwon (2024) shows that increased interactivity is associated with higher behavioral intentions to use among learners to interact with AI.

Not all the variables we tested showed a significant association with perceived interactivity of learner-AI interaction. Results did not show significant associations with perceived interactivity scores across ages, genders, education levels, and fields of education. We offer the findings as supplementary evidence of our scale's validity and utility. Specifically, our findings support that perceived interactivity of learner-AI interaction is not constrained by ages, genders, education levels, and fields of education. Perhaps this is because people generally have the ability to perceive interactivity in human-AI interaction (Sundar, 2020). Nevertheless, due to the limited numbers of participants in some categories (e.g., master's degree, doctor's degree, and the field of medicine), more research on the categories is expected in the future.

5 Limitations

While this study employed a systematic and step-by-step approach in developing and validating the scale, the following limitations should be acknowledged since they pave the way for future research. First, the participants were all university students, thus representative only of this sample scope. Although university students constitute a majority of learners utilizing AI in learning (Zhai et al., 2021), it is expected that data from diverse education levels (e.g., K-12 students) or learners pursuing professional development after graduation will further validate the findings of this study. Second, although the scale has been validated with a large sample size, learner distribution was not balanced as expected. Some categories were under-represented, such as the age above 26, master's degree, doctor's degree, the education fields of science and medicine, and the average duration of using AI lasting for 1–2 h and more than two hours. Third, while the item generation step incorporated a diverse group of participants from China, including undergraduate, master's, and doctoral students, the sample during the data collection was mainly composed of undergraduate students from China. Therefore, it is suggested that future research may recruit participant groups in similar ratios or even from other countries or cultures to generalize the results.

6 Theoretical and practical implications

This study provides insights into both theoretical and practical implications. Theoretically, the framework of perceived interactivity of learner-AI interaction that has been conceptualized in this study may be utilized to compare learners' perceived interactivity when using specific AI tools in terms of the four dimensions: responsiveness, personalization, learning engagement, and learner control. This scale provides an evaluation framework for performing comparative analyses across various AI tools in the four dimensions. Moreover, the scale may be adopted or adapted for research within specific contexts. In line with the literature (Memarian & Doleck, 2024; Petersen et al., 2022; Wang & Cheung, 2024), researchers may use the scale to extend scholarly investigation into relationships concerning the aspects closely related to learners' use of AI, such as motivated learning behaviors, trust, attitudes, self-efficacy, perceived ease of use, perceived usefulness, subsequent performance, and collaborative learning. Additionally, future research may use this scale to examine and develop hypotheses and theories on boundary conditions on perceived interactivity that may involve characteristics of learners and learning contexts (Liu & Shrum, 2002; Wang & Cheung, 2024), thereby deepening understandings regarding more effective ways of using AI in education. Finally, learners' perceived interactivity of learner-AI interaction might change over time as they accumulate experience of using AI in learning, so longitudinal studies are recommended to examine changes in perceived interactivity of learner-AI interaction and learning behaviors in different periods.

Practically, it is crucial to employ a multifaceted approach to analyzing perceived interactivity of human-AI interaction from the perspectives of AI practitioners, teachers, and institutional stakeholders. For AI practitioners, it is important to understand the four different dimensions of perceived interactivity when learners use various AI tools. Instead of focusing solely on the capabilities of AI tools, AI practitioners may use the scale to identify learners who do not perceive sufficient interactivity in one or multiple dimensions, and design the human-computer interface to address the gaps. For teachers, understanding students' interactions with AI and learning behaviors is necessary because teachers may adjust teaching strategies to leverage or mitigate the technological environment. Through the adjusted teaching strategies, students will get more assistance from teachers and be better prepared for the integration of AI into learning. For institutional stakeholders, the scale may be used for evaluating and choosing appropriate AI tools for specific learning objectives, and assessing students' learning experiences taken over a period of time. AI tools for learning are expected to expand in the future. This scale, specifically developed for the context of learners' use of AI, could reveal patterns of the learning process for institutional stakeholders' considerations to build constructive interactive learning environments.

7 Conclusion

This study is the first attempt to develop and validate the Perceived Interactivity of Learner-AI Interaction Scale to measure how learners perceive interactivity with AI. The study contributes to the existing literature by conceptualizing a framework of perceived interactivity of learner-AI interaction consisting of four dimensions, by developing a generic 17-item instrument for measuring perceived interactivity of learner-AI interaction, and by proposing potential applications of this instrument.

First, grounded in the Engagement Control Model, the study has conceptualized a framework of perceived interactivity of learner-AI interaction, including four dimensions: responsiveness, personalization, learner control, and learning engagement. The framework is based on the clearly defined concepts of interactivity and perceived interactivity, which were inconsistently defined and used in empirical studies (Petersen et al., 2022). The development of the scale pushes theories related to perceived interactivity of learner-AI interaction forward.

Second, the study has developed a generic 17-item instrument for measuring perceived interactivity of learner-AI interaction. The 17 items were developed following a rigorous scale refinement and validation process. Regarding validity, both the EFA and the CFA endorsed the instrument's validity, showing satisfactory results for each type of analysis. Regarding reliability, all four dimensions showed satisfactory psychometric properties, with four Cronbach's alpha values ranging from 0.820 to 0.915 and an overall Cronbach's alpha value of 0.948. It can be concluded that this scale is a valid and reliable instrument to measure learners' perceived interactivity of learner-AI interaction.

Third, this scale has many potential applications. In our initial applications, perceived interactivity of learner-AI interaction was significantly associated with learners' behavioral intentions to use AI in learning, months of using AI in learning, and average duration of using AI each time. Moreover, learners have demonstrated significant differences in their perceived interactivity of learner-AI interaction when using various AI tools, with intelligent tutoring systems scoring the highest, followed by personalized learning systems, generative AI, writing assistant, and automatic formative assessment. Additionally, this study did not find that perceived interactivity of learner-AI interaction was significantly associated with ages, genders, education levels, and fields of education. Future research may also consider exploring boundary conditions for perceived interactivity of learner-AI interaction related to characteristics of learners and learning contexts.

The findings of this study can be applied by scholars and educators in developing and testing theories that explain and predict learners' behaviors in using AI in learning, and by AI practitioners, teachers, and institutional stakeholders in applying AI tools in education. We hope this study will provide information on perceived interactivity, which is an important component of learning quality when learners use AI in learning, for scholars, educators, AI practitioners, teachers, and institutional stakeholders who consider the quality of education in human-AI interaction.

Author contribution (1) Feifei Wang – Conceived and designed the analysis, collected the data, performed the analysis, and wrote the manuscript. (2) Alan C. K. Cheung - Conceived and designed the analysis, col-

lected the data, supervised the study, and reviewed the manuscript. (3) Ching Sing Chai - Conceived and designed the analysis, planned methodology, and reviewed the manuscript. (4) Jin Liu – Collected the data, managed the data collection, and reviewed the manuscript.

Funding No funding was received for conducting this study.

Data availability The data used and analyzed in the study will be available on reasonable request.

Declarations

Ethical approval This study has received the ethics approval from The Chinese University of Hong Kong (Reference No. SBRE-23-0541).

Informed consent Informed consent has been obtained from all individuals included in this study.

Competing interests The authors have no relevant financial or non-financial interests to disclose.

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Authors and Affiliations

Feifei Wang[1] · Alan C. K. Cheung[1] · Ching Sing Chai[2] · Jin Liu[3]

✉ Alan C. K. Cheung
alancheung@cuhk.edu.hk

Feifei Wang
ffwang@link.cuhk.edu.hk

Ching Sing Chai
CSChai@cuhk.edu.hk

Jin Liu
liujin8899@outlook.com

[1] Department of Educational Administration and Policy, Faculty of Education, The Chinese University of Hong Kong, Hong Kong, China

[2] Department of Curriculum and Instruction, Faculty of Education, The Chinese University of Hong Kong, Hong Kong, China

[3] School of Education, Beijing Institute of Technology, Beijing, China